

Compared to a constant trading rate, the AIM adaptation tactic will incur a lower cost on average but will have increased risk exposure. The cost distribution of the AIM tactic is shown in Figure 8.3b. Notice that its expected cost C_3 is lower than that of the constant rate C_1 , but it does have higher bad risk exposure.

Compared to the targeted cost strategy the AIM tactic will trade at a slightly slower rate. Notice that the optimal AIM strategy is 1/3 of the optimal targeted cost solution. This results in a slightly lower cost than the targeted tactic and an increased potential for better prices if the favorable trend continues. But it is also associated with increased risk exposure and a potential for higher costs in times of adverse price movement. Figure 8.3c shows the cost distribution of the targeted cost tactic to the AIM tactic. Notice that the AIM cost C_3 is lower than the targeted cost C_2 but also has increased risk exposure as shown by the fatter tail on the right hand side (bad risk).

Passive-in-the-Money

The passive-in-the-money (PIM) adaptation tactic is a price-based scaling tactic intended to limit the potential losses and high costs in times of adverse price movement. It allows investors to better participate in gains to share in gains in times of favorable price movement.

The PIM tactic was originally designed with the Arrow-Pratt constant relative risk aversion (CRRA) formulation (see Pratt, 1964 and Arrow, 1971) in mind. But after conversations with a large number of investors we revised this adaptation tactic to be the mirror image of the AIM tactic. For example, investors may decelerate when trading prices are favorable because they believe the prices are going to continue to improve, and by trading slower they will further reduce market impact and continue to realize better prices. PIM will trade faster in times of adverse movement because investors believe that the adverse trend will continue to worsen. Thus we would rather pay higher market impact cost to avoid the most adverse prices and minimize the “bad” fat tail events. In other words, the PIM adaptation tactic minimizes potential bad outliers and increases our chances of achieving the good outliers.

The PIM tactic is found by maximizing the negative of the AIM adaptation tactic. Mathematically this formulated as:

$$\text{Max } L = - \frac{E_0(\text{Cost}) - E_t(\text{Cost})}{E_t(\text{TR})}$$